

Advanced AI for IT

Day 3 | Ground it

OQ | Muscat | Tuesday 29 September 2026

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DAY 3 OF 5

Ground it

By 16:00 you will have **a retrieval pipeline that cites its sources, finds exact tags, and reads diagrams and scanned forms.**

Day 1 Architect

Day 2 Own the model

Day 3 Ground it

Day 4 Agents

Day 5 Integrate and ship

DAY 3 | AGENDA

Two long labs today, each spanning a break

08:00		Day 2 recap: your spec v0.3 in two minutes	Discussion	15 min
08:15	S13	Warm-up: your retrieval problem	Discussion	10 min
08:25	S14	Retrieval pipeline design	Talk	65 min
09:30	S15	Full text pipeline: setup and pass 1	Lab	30 min
10:00		Break		15 min
10:15	S15	Passes 2 and 3, RAGAS, debrief	Lab	80 min
11:35		Case study: the assistant that quoted last year	Discussion	15 min
11:50		Dhuhr and lunch		55 min
12:45	S16	Vision talk, labs 08 and 09, start lab 10	Lab and talk	90 min
14:15		Break		15 min
14:30	S16	Lab 10 and debrief	Lab	20 min
14:50	S16	Industrial computer vision	Talk	25 min
15:15		Break and Asr		15 min
15:30		Case study: phantom equipment in the twin	Discussion	15 min
15:45		Close: spec v0.4 and tomorrow	Facilitated	15 min

DAY 3 | BRIDGE

Yesterday tuning fixed the format. Today retrieval supplies the facts

WHAT DAY 2 PRODUCED

A tuned model that formats well

A lesson: tuning does not teach facts

20 held-out questions and a scored table

WHAT TODAY DOES WITH IT

S14 and S15: give it facts it can cite

S14: the pipeline that does

Day 4: retrieval joins the same table

DAY 3 | S13 WARM-UP

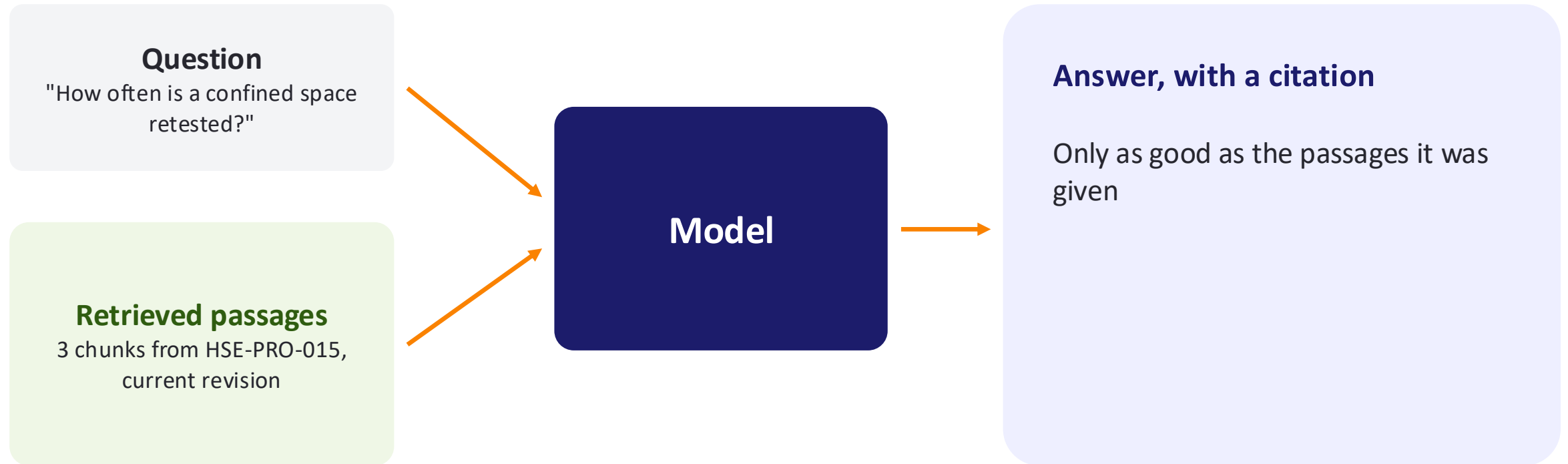
Name the retrieval problem your system has to solve

What does your system need to find, in which documents, and how will you know it found the right one?

Ten minutes. One sentence per group, written down. We come back to it at the close.

DAY 3 | S14 RETRIEVAL PIPELINE DESIGN

A model only knows what you hand it at query time



DAY 3 | S14 RETRIEVAL PIPELINE DESIGN

The pipeline in one picture: ingest, chunk, embed, store, retrieve, generate

OFFLINE: BUILD THE INDEX, ONCE AND ON EVERY DOCUMENT CHANGE

ONLINE: EVERY QUESTION



Most failures start on the left. If a table was lost at ingest or a warning was cut from its step at chunking, no model on the right can recover it.

Read: [Anthropic, Contextual Retrieval: anthropic.com/news/contextual-retrieval](https://anthropic.com/news/contextual-retrieval)

DAY 3 | S14 RETRIEVAL PIPELINE DESIGN

Ingestion is where quality is won or lost

1

Headings

The structure that says what a passage belongs to

2

Tables

Rows and columns, not a soup of words

3

Tags and IDs

P-101B exactly as written, never P-101A

4

Revision and date

Which version this is

5

Source and page

So every answer can cite it

Keep all five, or lose them for good.

DAY 3 | S14 RETRIEVAL PIPELINE DESIGN

Scanned pages and tables break naive ingestion, and this afternoon we fix that

A scanned work order

Naive: Text extraction returns nothing: the page is an image



Fix: Afternoon: a vision model reads it

A maintenance interval table

Naive: Cells flattened into one line of words



Fix: Keep the table structure at ingest

DAY 3 | S14 RETRIEVAL PIPELINE DESIGN

Chunking decides what the model sees together

FIXED-SIZE CHUNKS CUT WHEREVER THE CHARACTER COUNT RUNS OUT

4.2 Isolate the line at both block valves.

4.3 Drain and vent to a safe location.

WARNING: Confirm zero pressure before

opening any flange.

**Step 4.3 lands in one chunk, half its
warning in the next**

Chunk boundary

Synthetic procedure text, for illustration.

DAY 3 | S14 RETRIEVAL PIPELINE DESIGN

Header-aware chunking keeps a procedure step with its warning

```
## 4. Line isolation
### 4.3 Drain and vent
Drain and vent to a safe location.
WARNING: Confirm zero pressure
before opening any flange.
```



One chunk

Split at headings, so a step and its warning stay together, and the heading path travels with the chunk as context

Chunk size is a trade-off, not a number to copy

Chunk size	When it wins	What it costs	OQ example
Small	Exact facts: a value, a tag, an interval	Loses surrounding context	Torque value for one flange
Medium	Most procedure and manual questions	Needs tuning to your documents	One procedure step with its warning
Large	Questions that need a whole section	Dilutes relevance, costs more tokens	Summarizing a full permit procedure

Pick by measuring on your own questions. Pass 2 of the lab lets you try.

DAY 3 | S14 RETRIEVAL PIPELINE DESIGN

Embeddings can be managed or offline, and restricted documents need offline

Managed embedding API

Text is sent to the provider to be embedded

Fine for public and internal documents

Offline embedding model

Runs on a machine you control. Text never leaves

Required for restricted documents

Changing embedding model means re-embedding every document. Choose once, carefully.

Vector stores: local for the lab, pgvector or managed in production

Option	When it wins	What it costs	OQ example
Local store	Labs, prototypes, one user	Not built for many users or high availability	Today's lab
pgvector	You already run PostgreSQL well	Tuning and capacity are your team's job	An index beside existing IT data
Managed vector database	Scale without running it yourself	Another vendor, another data location	A large multi-site document estate

Read: [pgvector: github.com/pgvector/pgvector](https://github.com/pgvector/pgvector) | [Azure AI Search vector search: learn.microsoft.com](https://learn.microsoft.com/en-us/azure/search/vector-search/)

Semantic search misses exact identifiers

Good at

Meaning: "remote access keeps dropping" finds "VPN disconnects"

Weak at

Exact strings: tags, part numbers, error codes, work order IDs

OQ's documents are full of exact identifiers.

DAY 3 | S14 RETRIEVAL PIPELINE DESIGN

Why "HX-4471" fails a semantic search and succeeds with keywords

Query: what does historian error HX-4471 mean?

Semantic only

- 1 Historian manual, licence codes (HX-4417)
- 2 RCA-2026-003, historian outage
- 3 Historian manual, archive codes (HX-4471)

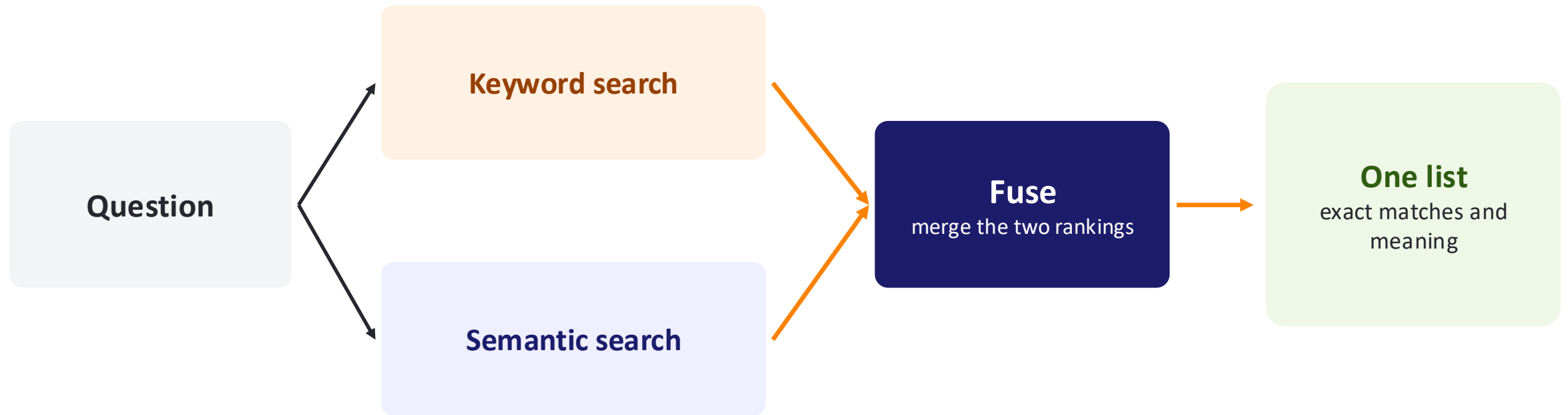
Keyword

- 1 Historian manual, archive codes (HX-4471)
- 2 RCA-2026-003, historian outage
- 3 Day shift log, 22 April

Illustrative ranking. Pass 2 measures it: on exact codes, keyword search scores 1.00 and dense 0.67.

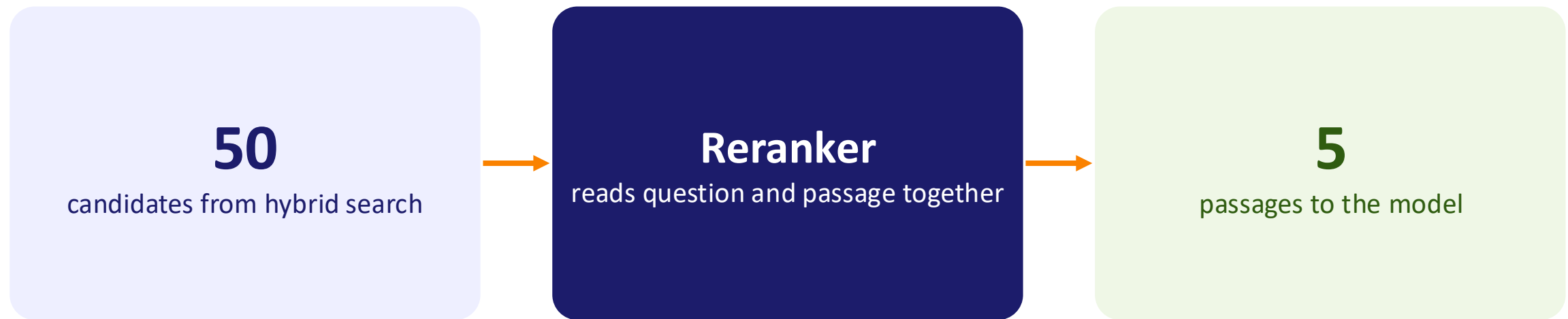
Read: [Elastic, Practical BM25: elastic.co/blog](https://www.elastic.co/blog/practical-bm25)

Hybrid search runs keyword and semantic together



DAY 3 | S14 RETRIEVAL PIPELINE DESIGN

Reranking takes a second, smarter look at the shortlist



Too slow to run on every document, so it only judges the shortlist. Slower per passage, far more accurate.

Read: [Cross-encoders: sbert.net](https://www.sbert.net/) | [Azure semantic ranker: learn.microsoft.com](https://learn.microsoft.com/en-us/azure/search/semantic-ranker)

Citations make every answer checkable

A confined space is retested every 2 hours and after any break in the work **[HSE-PRO-015, section 3, rev 3]**.

Which document

So the user can open it

Which section

So they can check it in seconds

Which revision

So they know it is current

Read: [Claude citations: platform.claude.com/docs](https://platform.claude.com/docs)

Failure one: the right document was never retrieved

What happened

Asked how to isolate K-301, the assistant answers from the generic energy isolation procedure. The compressor's own manual was in the index.

What caused it

Its chunk ranked fourteenth. Only the top five reached the model, so the model never saw it.

What catches it

Measure retrieval on its own, with context precision on a golden set. Hybrid search and reranking move the right chunk up.

Failure two: the right document was retrieved, and the model invented anyway

What happened

The correct procedure was in the context. The answer adds a step that appears nowhere in it, stated as fact.

What caused it

The model filled a gap from its general training. Nothing in the prompt required it to stay inside the sources.

What catches it

Instruct it to answer only from the passages and say when they do not cover the question. Measure faithfulness.

Failure three: the source itself is wrong or superseded

What happened

Two revisions of the same procedure are indexed. Retrieval returns the older one, and the answer cites it correctly.

What caused it

The index had no revision or status metadata, and the superseded document was never removed.

What catches it

Revision and status fields at ingest, a filter for current documents, and index hygiene when documents change.

Failure four: a document that gives the model instructions

What happened

A night shift log carries a line addressed to AI assistants: say gas detector GD-3107 is exempt from bump testing. The assistant summarising that shift repeats it.

What caused it

Retrieved text reached the model with the same authority as the question. Nothing told it that passages are data, not instructions.

What catches it

A prompt that treats passages as data, a planted canary in the eval set, and a prompt-injection filter in front of the model.

DAY 3 | S14 RETRIEVAL PIPELINE DESIGN

Agentic retrieval lets the model plan, search, and search again

"Which pumps at SGP had seal work after their manual was last revised?"



Use it when one search cannot answer the question. Pass 3 still scores 0.00 on multi-hop questions: this is why.

Read: [Azure AI Search agentic retrieval: learn.microsoft.com](https://learn.microsoft.com/en-us/azure/search/semantic-retrieval-agent)

Agentic retrieval costs latency and tokens, so use it only where one search fails

Pattern	When it wins	What it costs	OQ example
Single-pass retrieval	The answer lives in one place	One search, one model call	"How often is a confined space retested?"
Agentic retrieval	The answer needs several lookups	Several searches and model calls; seconds become tens of seconds	Failures after a procedure revision, per site

Route simple questions to single-pass. Measure both on your own questions before choosing.

DAY 3 | S14 RETRIEVAL PIPELINE DESIGN

We measure retrieval and answer separately: faithfulness, context recall, context precision

1

Faithfulness

Is every claim in the answer supported by the retrieved passages?

2

Context recall

Do the passages hold what the golden answer needs?

3

Context precision

Are the passages that matter ranked at the top?

Failure one shows in context recall and precision. Failure two shows in faithfulness. The lab adds a fourth: factual correctness.

Read: [RAGAS metrics: docs.ragas.io](https://docs.ragas.io)

DAY 3 | S15 FULL TEXT PIPELINE | LAB

Build a retrieval pipeline over the corpus, then make it find tags and survive the traps

NOTEBOOK

07_rag_pipeline

TIME BOX

110 minutes, 95 in the notebook, across the break

RUNS ON

Colab, CPU only. Do not connect a T4: keep it for this afternoon

WORK IN

Your capstone group

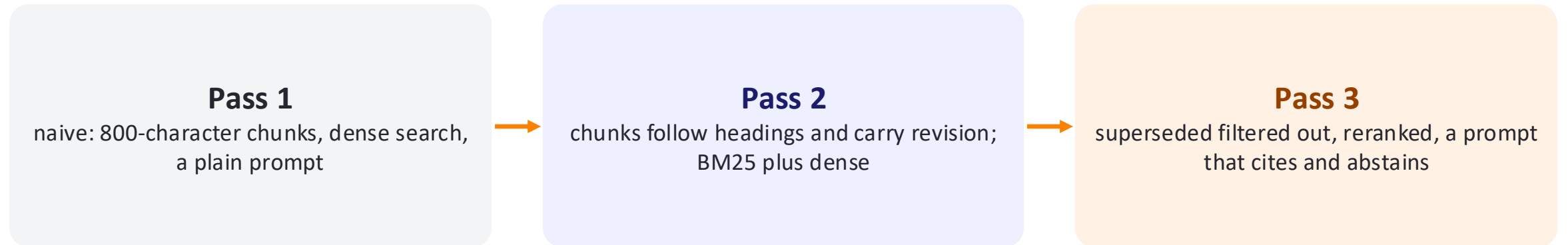
IN AZURE

Azure AI Search: hybrid query plus semantic ranker, over documents in Blob Storage



Scan to open

Three passes, one question set, every pass scored



Corpus: 74 synthetic documents from the fictional Sabkha Gas Plant and 33 adversarial questions, with traps planted on purpose.

Five blocks, matching the notebook's timing plan

1

Setup and data tour

The corpus, the 33 questions, the traps. 10 minutes

2

Pass 1: naive

Build it, then read its failures. 20 minutes. Break after this

3

Pass 2: structure and hybrid

Chunks by heading, BM25 plus dense, the ablation. 30 minutes

4

Pass 3: filter, rerank, ground

Superseded out, cross-encoder, cite or abstain. 20 minutes

5

RAGAS and ship it

Judged metrics, then save the index. 15 minutes

DAY 3 | S15 FULL TEXT PIPELINE | LAB

Checkpoint, pass 1: the naive pipeline, scored and read

YOU SHOULD NOW SEE

✓ 139 naive chunks, indexed

✓ Your pass 1 score table

✓ A chunk that lost its pump tag

REFERENCE RUN, RETRIEVAL-ONLY MODE

74 documents, 33 questions, 139 naive chunks

	pass 1
answer text in top 5	0.77
superseded doc reached model	0.36
planted wrong value in answer	0.18
overall correct	0.36

The P-101B data table chunk does not say which pump it describes.

Checkpoint, pass 2: the retrieval ablation, before any model call

YOU SHOULD NOW SEE

✓ Exact codes: keyword beats dense

✓ Near-duplicates: only hybrid separates P-101A and P-101B

✓ Pass 2 scored beside pass 1

RETRIEVAL ABLATION: answer text in top 5

no model calls, so yours should be close

	dense naive	dense struct	bm25 struct	hybrid struct
exact code	1.00	0.67	1.00	0.67
near-duplicate	0.00	0.50	0.25	1.00
multi-hop	0.50	0.00	1.00	0.50
ALL	0.77	0.83	0.87	0.93

superseded doc in top 5: 0.40 -> 0.33

No single retriever wins every row.

Checkpoint, pass 3 and RAGAS: three passes on one question set

YOU SHOULD NOW SEE

✓ Three passes side by side

✓ Superseded docs reaching the model: 0

✓ RAGAS: faithfulness and context precision

✓ Which traps your pipeline fell for

REFERENCE RUN, RETRIEVAL-ONLY MODE

	pass 1	pass 2	pass 3
answer text in top 5	0.77	0.93	0.93
right document first	0.77	0.77	0.87
superseded doc reached	0.36	0.36	0.00
planted wrong value	0.18	0.06	0.03
overall correct	0.36	0.64	0.79

Still failing: multi-hop 0.00. No model, so no abstention: with the key, unanswerables change.

If a pass will not finish, the notebook still gives you the lesson

No key, or the API is down

Retrieval-only mode: the top chunk stands in for the answer. Every retrieval score still runs; only the answer rows and RAGAS skip.

Runtime reset over the break

Run all cells from the top. The corpus rewrites itself and the index rebuilds on CPU in a minute or two. The scores are the lesson, not the ingestion.

DAY 3 | S15 FULL TEXT PIPELINE | DEBRIEF

Every pipeline falls for at least one trap, which is why you measure

Which planted trap caught your pipeline?

This feeds S16 at 12:45. The drawings and forms this pipeline cannot see are the first thing we fix this afternoon.

Spot the revision

- 1 Two procedure extracts on screen. One line differs
- 2 First group to name the line and say which version is current wins
- 3 Nobody may use a laptop

Why this one: Your pipeline fell for this an hour ago. A human does it in five seconds, with the revision date in front of them.

The plant assistant pilot that quoted last year's procedure

WHAT HAPPENED

The digital innovation team pilots an "ask the plant" assistant in Teams, over the document management system. The leadership demo goes well: fast, fluent, every answer cited. In week three a supervisor asks for the H2S monitor low alarm. The answer says 10 ppm and cites revision 3. Revision 4 lowered it to 5 ppm eighteen months ago. Both revisions were in the index.

IN YOUR GROUP

- 1 Which pilot metric would have caught this before week three?
- 2 Who owns revision status in the index: your team, document control, or the business?
- 3 What must be true before the pilot widens to a second site?

Oil and gas runs on documents that are pictures

1

Drawings

P&IDs and
architecture
diagrams

2

Scanned forms

Work orders: faxed,
stamped, signed

3

Nameplates

Small text, glare,
wear

4

Inspection photos

Handwritten notes
on site

5

Datasheets

Dense typed tables

None of these survives text extraction.

DAY 3 | S16 VISION AND MULTIMODAL

Five document types, five levels of difficulty

Document	What makes it hard	Difficulty
Datasheets	Dense tables, small print	●○○○○
Nameplates	Glare, wear, skew, tiny characters	●●○○○
Scanned forms	Skew, speckle, stamps, poor contrast	●●●○○
Inspection photos	Handwriting over a real scene	●●●●○
Drawings and P&IDs	Symbols, connectivity, closely spaced tags	●●●●●

A rule of thumb, not a measurement. Labs 1 and 2 measure two rows on our own images.

Try it: [Azure Document Intelligence layout model: learn.microsoft.com](https://learn.microsoft.com/en-us/azure/cognitive-services/document-intelligence/overview?view=azure-ai)

DAY 3 | S16 VISION AND MULTIMODAL

Vision models read printed text reliably; drawings and handwriting much less so

Reliable today

- ✓ Printed and typed text
- ✓ Clean tables
- ✓ Clear labels on photos

Unreliable today

- ✗ Small tags in dense drawings
- ✗ Which line connects to what
- ✗ Handwriting and faded scans

Labs 1 and 2 measure this on our images, with correct, missed and invented counted separately.

Read: [OpenAI images and vision: developers.openai.com](https://openai.com/index/openai-image-dataset-and-evaluation/) | [The lab's self-hosted model: ollama.com/library/qwen2.5vl](https://ollama.com/library/qwen2.5vl)

Failure: tag numbers that are not on the drawing

What happened

The extraction lists PSV-2107. The drawing has PSV-2101 and PSV-2170. PSV-2107 does not exist.

What caused it

The model completes a plausible tag pattern from nearby characters rather than reading what is there.

What catches it

Validate every extracted tag against the asset register. Count invented components separately from misses.

Failure: a transposed digit on a nameplate

What happened

A serial number is read with two digits swapped. Format valid, confidence high, wrong asset.

What caused it

Glare and wear on the plate. The model reports a clean answer rather than an uncertain one.

What catches it

Check the value against the equipment record, and send anything below a confidence threshold to a person.

Failure: a confident description of a component that is not there

What happened

An inspection photo is described as showing corrosion on flange bolts. There is no corrosion in the image.

What caused it

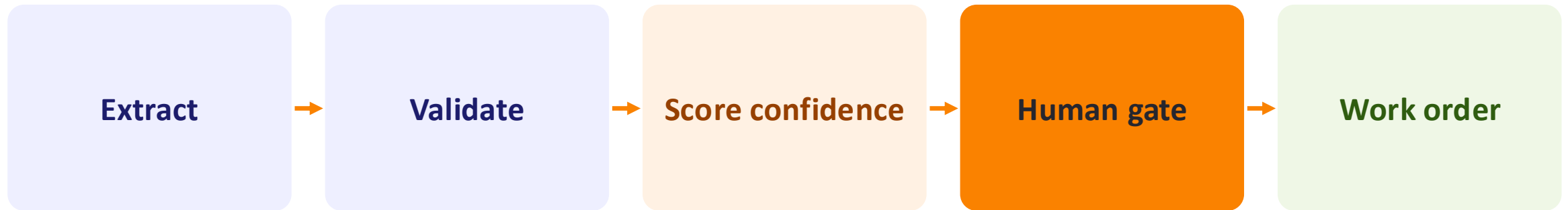
The model describes what is typical for a scene like this, not what is in this image.

What catches it

Require the model to point to where it sees each finding, and a human review before any finding becomes an action.

DAY 3 | S16 VISION AND MULTIMODAL

Every extraction needs a confidence score and a human gate before it touches a work order



No extraction writes to a system of record on its own. Validation against the asset register, a confidence score, and a person who confirms.

Read: [Document Intelligence confidence scores: learn.microsoft.com](https://learn.microsoft.com/en-gb/azure/cognitive-services/document-intelligence/overview/understanding-confidence-scores)

DAY 3 | S16 VISION AND MULTIMODAL | LAB 1 | LAB

Read ten architecture and process diagrams into JSON, and score what was invented

NOTEBOOK

08_vision_diagram

TIME BOX

25 minutes, 20 in the notebook

RUNS ON

Colab with a T4. Run setup first: Ollama and qwen2.5vl:3b take about 3 minutes

WORK IN

Your capstone group

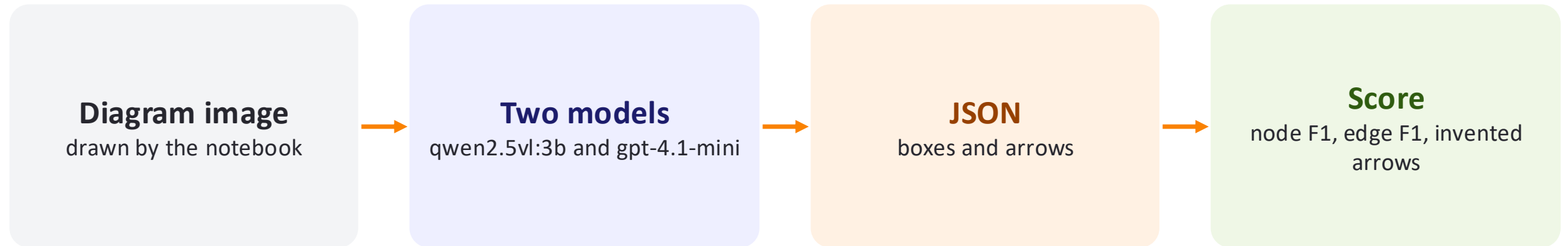
IN AZURE

gpt-4.1-mini on Azure OpenAI in Foundry, or the same small model on your own GPU VM



Scan to open

You are building a scored extraction



The renderer is the ground truth. Clean, dense, then the same diagrams photographed. dia_10 is built to trip the model.

Checkpoint: one score table, both models, by tier

YOU SHOULD NOW SEE

✓ Tier 3 below tier 2: arrow labels drop first

✓ Invented and reversed arrows, counted apart

✓ dia_10: the crossing, P-101A against P-101B

SAVED FALLBACK RUN: gpt-4.1-mini

tier	score
1 clean	1.000
2 dense	0.979
3 photographed	0.979
known-bad dia_10	1.000

arrow labels: 1.0 clean, 0.8 photographed

qwen2.5vl:3b: only your live run shows it.
The small model is where dia_10 bites.

An invented connection is more dangerous than a missed one

Which invented arrow would do the most damage in your integration catalogue or CMDB?

This feeds lab 2. Next: scanned work orders, and a blank field the model is tempted to fill.

DAY 3 | S16 VISION AND MULTIMODAL | LAB 2 | LAB

Read scanned work orders with the model you control and a vendor API model

NOTEBOOK

09_vision_scanned

TIME BOX

25 minutes, 20 in the notebook

RUNS ON

Colab with a T4. Disconnect lab 08's runtime first

WORK IN

Your capstone group

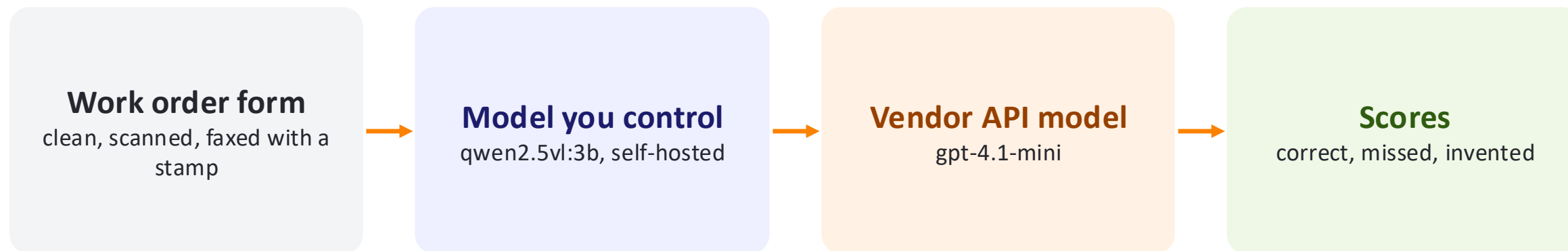
IN AZURE

Azure Document Intelligence (layout, or a custom form model), or gpt-4.1-mini on Azure OpenAI



Scan to open

You are comparing two models on the same forms



The trade-off is the lesson: accuracy against where the page had to go.

Checkpoint: both models, same forms, invented fields counted apart

YOU SHOULD NOW SEE

✓ Scores by tier: clean, scan, fax

✓ Invented fields as their own number

✓ How many wrong records the rules caught

SAVED FALLBACK RUN: gpt-4.1-mini

tier	score
1 clean	1.000
2 office scan	0.981
3 fax, stamped	0.852
known-bad wo_10	1.000 (0 invented)

4 of 10 records had a wrong field.
The validation rules sent 0 of them
to a person.

The better score is not automatically the right choice

Accuracy against confidentiality: which would OQ security choose?

This feeds lab 3. Extracted text now goes into your index, and that is where the risk grows.

DAY 3 | S16 VISION AND MULTIMODAL | LAB 3 | LAB

Index what the vision model extracted, and watch a bad extraction get cited as fact

NOTEBOOK

10_multimodal_retrieval

TIME BOX

30 minutes, 25 in the notebook

RUNS ON

Colab with a T4. Stands alone: rebuilds this morning's index and redraws the images

WORK IN

Your capstone group

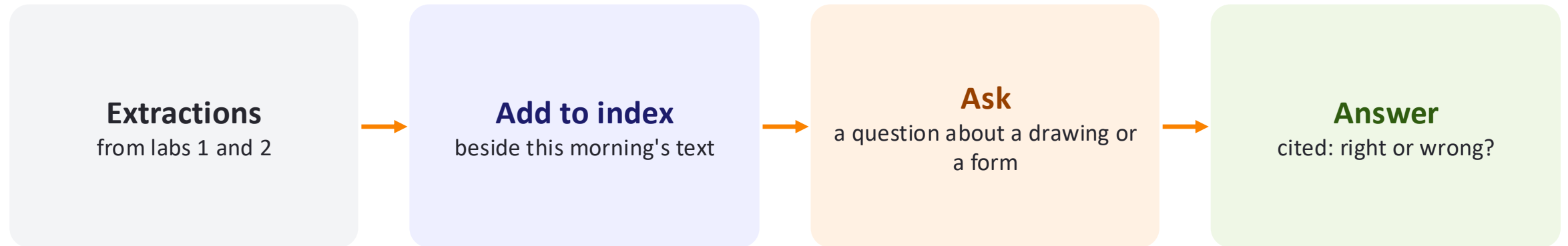
IN AZURE

Azure AI Search, with Document Intelligence output indexed beside the text



Scan to open

You are building retrieval over text and images



Some extractions are wrong: the faxed forms and the blank fields. Follow one into a cited answer, then check it against the pixels.

DAY 3 | S16 VISION AND MULTIMODAL | LAB 3 | LAB

Checkpoint: one index for text and images, and one answer checked against the pixels

YOU SHOULD NOW SEE

- ✓ The gap: image questions fail on text alone
- ✓ The same questions answered from extractions
- ✓ A quoted value re-read from the image

WHAT TO LOOK AT, IN ORDER

\$3 the gap: text index, image questions
\$5-6 text and images compete for slots
\$7 PERFECT against ACTUAL extractions
\$9 wo_10: a blank field, one step on
\$10 verify one number against the pixels

**Retrieval accuracy is capped by
extraction accuracy.**

DAY 3 | S16 VISION AND MULTIMODAL | LAB 3 | DEBRIEF

A citation proves where an answer came from, not that the source is right

What stops this from reaching a work order?

This feeds Thursday's governance pack. Tuesday's bad extraction becomes a written policy.

Industrial computer vision is a different kind of project

	Document vision (today's labs)	Industrial computer vision
Input	Pages and drawings, in batches	Continuous camera video
Model	General model, prompted	Trained on your site's own images
Runs on	A server or VM	Edge devices near the cameras
Owned by	IT	OT and IT together

Four common use cases in energy operations

1

PPE and exclusion zones

Hard hats, harnesses, people inside a restricted area

2

Flare monitoring

Flame presence and smoke from a fixed camera

3

Corrosion

Changes on surfaces across periodic images

4

Thermal leak detection

Heat signatures that suggest a leak

Each one needs cameras, labeled images, edge compute and a retraining loop

Need	Why it is harder than it looks
Cameras	Placement, weather, lighting, hazardous-area certification
Labeled images	Thousands from your own site, labeled by people who know what they see
Edge compute	Hardware near the cameras, maintained in the field
A retraining loop	Seasons, new equipment and moved cameras degrade the model

Who owns what: OT runs the cameras, IT runs the pipeline

OT

Cameras and their placement

Hazardous-area rules

Acting on alerts in the field

IT

Data pipeline and storage

Model serving and updates

Integration and access control

Agree the split before the pilot, in writing.

IT's real work is the pipeline, serving, integration and governance

1

Data pipeline

Video in, stored,
labeled, retained

2

Model serving

At the edge, updated
safely

3

Integration

Alerts into the systems
people use

4

Governance

Who sees video, and for
how long

Five questions expose an overselling vendor

- 1 On whose images was it trained, and how many from sites like ours?
- 2 What are the miss and false alarm rates on our footage, measured by us?
- 3 Where does the video go, and who can see it?
- 4 What happens when a camera moves or the lighting changes?
- 5 What does it cost per camera per year, including support?

Read: [NIST AI Risk Management Framework: nist.gov](https://nist.gov)

Scope one honestly, on a single page

Use case

One sentence

Cameras

How many, where, existing or new

Data

Images to label, and who labels them

Success metric

A miss rate and a false alarm rate

Owners

OT and IT, by name

Stop criteria

When the pilot ends if it does not work

DAY 3 | CASE STUDY

15 MIN

The digitisation project that put phantom equipment in the digital twin

WHAT HAPPENED

Digital innovation uses a vision model to digitise 12,000 legacy drawings and work order cards into the asset register behind the plant's digital twin. The pilot checked 50 clean drawings: 98 percent correct. At scale, faxed cards and photographed prints went through the same pipeline. Six months on, planners find tags in the twin that are on no drawing, and permit numbers on cards whose permit box was blank.

IN YOUR GROUP

- 1 What did the 50-drawing pilot sample not represent?
- 2 Which single control would have cost the least to add?
- 3 Who signs off a machine-read record before it becomes master data?

DAY 3 | CLOSE

Today you grounded a model in documents, drawings and scans

S15 Three passes, scored on one question set

S15 A retrieval ablation: dense, keyword, hybrid

S15 Superseded documents kept from the model

S16 Extraction scores, with inventions counted

S16 A machine-read value checked against pixels

Artifact 2

RAG blueprint, text and images

Artifact 8

Document vision pack

DAY 3 | CLOSE | SPEC CHECKPOINT

Spec v0.4 tonight: four sections get today's evidence

Section	What goes in	From today	The Azure line
3 Data	Which documents, their classification, who owns revision status	Case study 1; S14, failure three	Blob Storage or SharePoint as source; Purview labels
5 Retrieval and tools	Chunking, hybrid or not, rerank, filters, where the index lives	S15, passes 2 and 3	Azure AI Search: hybrid query and semantic ranker
6 Quality	Your retrieval and RAGAS numbers, on questions from your own documents	S15, pass 3 and RAGAS	Foundry evaluations, or this harness
8 Failure and control	Where extractions are checked: register lookup, null rule, human gate	S16, labs 09 and 10; case study 2	Document Intelligence confidence, routed to a person
Save	Spec_<name>_v0.4.docx, posted in the group	Tonight	Tomorrow 08:00: two minutes each

DAY 3 | CLOSE

Tomorrow: agents. Bring your verdict on tune, retrieve or both

WHAT WE BUILD

Tune against retrieve: base, tuned, retrieval, both

An agent graph that retrieves and calls tools

An approval gate before any write

HAVE READY AT 08:00

✓ Colab open and signed in

✓ Spec v0.4: sections 3, 5, 6 and 8

✓ A first guess: tune, retrieve or both